

CURRICULUM VITAE

Yunkai Yang

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EDUCATION

City University of Hong Kong, Hong Kong SAR

Sep 2026 - Jun 2027 (expected)

M.Sc. student in Neuroscience (expected)

Harvard University (Online)

Sep 2024 - Jan 2025

Evolutionary Psychology; Instructor: Max Krasnow; Final grade: 93.25/100

Henan University, Kaifeng, China

2021 - 2025

B.S. in Applied Psychology, Weighted average: 86.6/100

Thesis: Brain Mechanisms of Emotion Regulation from an Interpersonal Interaction Perspective

RESEARCH GRANTS AND GRANT-SUPPORTED RESEARCH

Henan University Student Innovation and Entrepreneurship Training Program

2023 - 2024

Role: Principal Investigator

Project: Mechanisms Underlying the Effects of Interpersonal Contact on Emotion

Amount awarded: CNY 10,000

Grant-Supported Research Reported in [1] and [6]

2023 - 2026

Sources: Youth Project of Humanities and Social Sciences Research, Ministry of Education; National Philosophy and Social Sciences Project; Key Project in Basic Research of Philosophy and Social Sciences in Higher Education Institutions of Henan Province

Role: Principal Investigator

Research expenditure: Approximately CNY 12,000 for the studies reported in [1] and [6]

Grant-Supported Research Reported in [2]

2022 - 2025

Source: Youth Project of Humanities and Social Sciences Research, Ministry of Education

Role: Principal Investigator

Research expenditure: Approximately CNY 15,000 for the study reported in [2]

PUBLICATIONS

1. Yang, Y., He, J., Wang, L., Wang, J., Xu, J., Jiang, X., Ao, N., Zhang, P., Chen, Y., Wang, M., & Du, F. (2026). Bidirectional benefits: Interbrain synchronization and role-specific neural signatures in interpersonal emotion regulation. *Emotion*. Advance online publication. <https://doi.org/10.1037/emo0001683>
2. Chen, Y., Niu, H., Ao, N., Xu, J., Ming, Q., Teng, X., Yang, B., Lian, Z., Jiang, X., & Yang, Y. (corresponding author) (2025). An fNIRS hyperscanning dataset on the modulation of synchrony by social relationship during social touch. *Scientific Data*. 12, 1975. <https://doi.org/10.1038/s41597-025-06233-9>
3. Huang, Y., Chen, X., Huang, J., Chen, B., Wang, W., Gao, T., Xing, Z., Liang, G., Peng, T., Huang, Y., Miao, Y., Wang, R., He, C., Li, L., Liu, Z., Zhang, P., Yang, Y. (corresponding author), & He, Z. (2026). Differential emotional responses and relief mechanisms across social exclusion roles. *Cognitive, Affective, & Behavioral Neuroscience*. Advance online publication. <https://doi.org/10.3758/s13415-026-01414-4>
4. Buchanan, E. M., Cuccolo, K., Heyman, T., van Berkel, N., Coles, N. A., Iyer, A., Peters, K., van 't Veer, A. E., Montefinese, M., Maxwell, N. P., Taylor, J. E., Valentine, K. D., Arriaga, P., Barzykowski, K., Boucher, L., Collins, W. M., Vaidis, D. C., Aczel, B., Al-Hoorie, A. H., ... Yang, Y., ... Lewis, S. C. (2025). Measuring the semantic priming effect across many languages. *Nature Human Behaviour*. 1-20. <https://doi.org/10.1038/s41562-025-02254-x>

5. **Yang, Y.** (co-first author), Zhao, Y. (co-first author), Xie, K., Song, C., Huang, Y., Kong, A., Huang, S., Xing, Z., Wang, W., Muhlert, N., Elliott, R., Tan, S., He, Z., & Zhang, D. (2025). Multi-timescale neural adaptation failure as a mechanistic signature of major depression. *bioRxiv*. [Preprint]. <https://doi.org/10.64898/2025.12.16.694510> Manuscript under major revision at *Communications Biology*.
6. **Yang, Y.**, Chen, Y., & Du, F. (n.d.). Neural decoding in social interaction: Hierarchical separation of intra- and interbrain features for representing process versus outcome information. [Manuscript under review at *Acta Psychologica Sinica*].
7. Kong, A., Wu, Z., Liu, F., Ni, J., Deng, J., Zhang, P., Huang, S., Hu, Y., Wang, Z., Yu, X., Li, Y., Hu, X., Zhu, H., Li, B., Jiang, Z., He, Z., Muhlert, N., **Yang, Y. (corresponding author)**, & Elliott, R. (2025). Economic deprivation, diminished social support, and loneliness: An empirical test of the EDSL model using UK Biobank data. *Research Square*. [Preprint]. <https://doi.org/10.21203/rs.3.rs-7553119/v1> Under review at *BMC Public Health*.
8. Chen, Y., Ao, N., Jiang, X., Niu, H., Xu, J., **Yang, Y.**, Du, F., & Wang, M. (2025). Interbrain synchrony during interpersonal touch is associated with the remission of negative emotion. *PsyArXiv*. [Preprint]. <https://doi.org/10.31234/osf.io/x76qz> Manuscript under minor revision at *BrainX*.
9. Jiang, X., Chen, Y., Pal, N. R., Chang, Y.-C., **Yang, Y.**, Do, T., & Lin, C.-T. (2025). Interpretable dual-filter fuzzy neural networks for affective brain-computer interfaces. *arXiv*. arXiv:2502.17445. <https://doi.org/10.48550/arXiv.2502.17445> Under review at *IEEE Transactions on Emerging Topics in Computational Intelligence*.

CONFERENCE PRESENTATIONS

10. **Yang, Y.**, He, J., Wang, L., Wang, J., Xu, J., Jiang, X., Ao, N., Zhang, P., Chen, Y., Wang, M., & Du, F. (2026). Bidirectional benefits: Interbrain synchronization and role-specific neural signatures in interpersonal emotion regulation. Poster presented at the International Society for Research on Emotion 2026 Conference, Brisbane, Australia.
11. **Yang, Y.**, Chen, Y., & Du, F. (n.d.). Neural decoding in social interaction: Hierarchical separation of intra- and interbrain features for representing process versus outcome information. Poster presented at the Annual Meeting of the China Association for Science and Technology, Beijing, China.
12. Si, B., Ma, Q., **YunKai Yang (co-first author)**, & Li, Z. (n.d.). Prediction of daily tourist volume in Hangzhou based on Baidu Index. Paper presented at the 8th International Conference on Economic Management and Green Development.

RESEARCH EXPERIENCE

Note. Bracketed numbers refer to items in the publication, manuscript, and conference sections.

Research Assistant, Shenzhen University, Shenzhen, China

2024 - Present

Supervisor: Prof. Zhenhong He | Appointment: part-time (2024-2025); full-time (2025-Present)

Project: *Multi-Timescale Neural Adaptation Failure as a Mechanistic Signature of Major Depression [5]*

- **EEG preprocessing:** Filtered continuous EEG, ran independent component analysis, interpolated bad channels, rejected artifact-contaminated segments, and epoched trials.
- **EEG analysis:** Conducted time-domain, frequency-domain, time-frequency, multivariate pattern analysis (MVPA), traveling-wave, and machine-learning analyses.
- **Visualization and writing:** Produced behavioral and EEG figures in R and Python; drafted the initial manuscript and coordinated revisions.
- **Output:** First-authored preprint [5]; the manuscript is under major revision at *Communications Biology*.

Project: *Differential Emotional Responses and Relief Mechanisms Across Social Exclusion Roles [3]*

- **Data collection:** Collected behavioral and EEG data across three experiments examining transitions between social-exclusion roles.

- **EEG and behavioral analysis:** Preprocessed EEG and conducted time-domain analysis and MVPA; analyzed behavioral data using ANOVA and correlation analyses.
- **Result:** Identified a reduction in the late positive potential during the excluded-to-excluder switch and a positive association between emotional relief and revenge.
- **Output:** Produced figures, drafted and revised the manuscript, and served as corresponding author on [3].

Project: Economic Deprivation, Diminished Social Support, and Loneliness [7]

- **Data preparation and analysis:** Prepared UK Biobank data and conducted machine-learning classification and structural equation modeling to test the EDSL model.
- **Visualization and writing:** Created R and Python visualizations and drafted the initial manuscript.
- **Output:** Served as corresponding author on preprint [7], currently under review at BMC Public Health.

Project: Dynamic Neural Deactivation Bridges Direct and Competitive Inhibition Processes

- **Data collection:** Collected cohort EEG and fMRI data.
- **EEG analysis:** Preprocessed EEG and conducted change-point detection, time-domain analysis, MVPA, and traveling-wave analysis.
- **fMRI analysis:** Implemented dynamic causal modeling, psychophysiological interaction analysis, first- and second-level models, and MVPA.
- **Output:** Produced behavioral, EEG, and fMRI figures and contributed to drafting and revision; the project was subsequently published in Advanced Science.

Project: Additional Projects in Emotion Regulation and Neurodevelopment

- **Imaginal reversal:** Used PsychoPy to collect data across nine online behavioral experiments; cleaned data and conducted ANOVA, mediation, and linear mixed-effects analyses; drafted the initial manuscript.
- **Autism neurodevelopment:** Analyzed resting-state fNIRS functional connectivity in children with autism.
- **Neuromodulation and emotion regulation:** Analyzed TMS data and modeled fMRI data using hidden Markov models, CNNs, Transformers, and neural-manifold approaches.

Undergraduate Research Assistant, Henan University, Kaifeng, China

2021 - 2025

Supervisor: Prof. Yanan Chen

Project: Interpersonal Emotion Regulation and Neural Decoding in Social Interaction [1, 6, 10, 11]

- **Experimental design and data collection:** Adapted dyadic fNIRS hyperscanning paradigms, implemented tasks in PsychoPy, and collected behavioral and fNIRS data.
- **fNIRS preprocessing and analysis:** Applied filtering, bad-channel rejection, negative-correlation enhancement, first- and second-level modeling, interbrain synchrony analysis, and MVPA.
- **Follow-up neural decoding:** Expanded the sample to 40 dyads by recruiting six additional dyads and used MVPA and representational similarity analysis to compare intra- and interbrain features. The analyses indicated that process information was primarily encoded within individual brains, whereas outcome information relied more strongly on interbrain coupling.
- **Outputs:** First-authored publications/manuscripts [1, 6] and presented the work at conferences [10, 11]; drafted the manuscripts and completed multiple rounds of revision.

Project: Social Touch, Social Relationship, and Interbrain Synchrony [2, 8]

- **Dataset development:** Collected fNIRS hyperscanning and behavioral-rating data from 47 female dyads (24 friend pairs and 23 stranger pairs) under touch and no-touch conditions.
- **Preprocessing and validation:** Preprocessed fNIRS signals, conducted technical validation and preliminary interbrain synchrony analyses, and prepared the open dataset and documentation.
- **Outputs:** Served as corresponding author on the Scientific Data article [2] and contributed to preprint/manuscript [8].

Project: Measuring the Semantic Priming Effect Across Many Languages [4]

- **Materials and data collection:** Prepared stimulus materials and collected behavioral rating data for a large multilingual collaboration.
- **Output:** Reviewed and revised the manuscript published in Nature Human Behaviour [4].

Project: Affective Brain-Computer Interfaces [9]

- **Data and analysis:** Collected fNIRS data and conducted preliminary analyses for an interpretable dual-filter fuzzy neural-network framework.
- **Output:** Drafted the initial manuscript and contributed to preprint [9].

Project: Mental Health Survey of Secondary-School Students

- **Data collection:** Collected questionnaire data for a school-based mental health screening project.
- **Data analysis:** Conducted latent profile analysis, structural equation modeling, machine learning, and network analysis.

SKILLS

Statistical and computational methods: Multiple regression; ANOVA; mediation and moderation; structural equation modeling; latent class/profile analysis; linear mixed-effects models; graph/network analysis; machine learning

Neural data analysis: ERP/time-domain, frequency-domain, and time-frequency EEG analysis; traveling-wave analysis; Multi-voxel pattern analysis (MVPA); representational similarity analysis (RSA); static, dynamic, and effective connectivity; interbrain synchrony; hidden Markov modeling; DCM; PPI

Programming and software: Python (MNE, fMRIPrep, NiMARE), R, MATLAB (SPM, CONN), SPSS, Mplus, LaTeX

Experimental platforms: PsychoPy, E-Prime, Psychtoolbox, Tobii, Curry 7

Data acquisition: Eye tracking (Tobii); fNIRS (NIRx); EEG (NeuroScan, Brain Products); fMRI (Siemens Prisma 3.0T and 7.0T)

Languages: Chinese (native); English (proficient)

Certification: Teacher Qualification Certificate (Senior High School Psychology)

SCHOLARSHIPS AND AWARDS

Travel Award, International Society for Research on Emotion 2026 Conference	2026
Outstanding Student Leader, Henan University	2023
Henan University Scholarship	2023
Provincial Project Approval, Henan University Innovation Program	2023
University Award, National Student Statistical Modeling Competition	2024
First Prize (University Level), National Student Innovation & Entrepreneurship Competition	2023

MEMBERSHIPS

International Society for Research on Emotion (ISRE)	2025 - Present
Psychological Science Accelerator	2022 - Present